

Reinforcement Learning Based Adaptive Traffic Signal Control Using Deep Q-Network Variants

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Abstract—Traffic congestion in urban areas leads to significant economic losses in terms of travel delays, air pollution, and unplanned urban growth, while conventional fixed-time or sensor-based traffic signal systems lack the flexibility required for real-time adaptation. In this paper, deep reinforcement learning (DRL) is investigated for adaptive traffic signal control by formulating the intersection management problem as a Markov decision process, in which an agent learns optimal signal phase decisions through interaction with a simulated environment. A realistic simulation framework is developed using the Simulation of Urban Mobility (SUMO) platform and the Traffic Control Interface (TraCI). Three DRL models are evaluated, namely Deep Q-Network (DQN), Double Deep Q-Network (DDQN), and Dueling Double Deep Q-Network (D3QN). The experimental results indicate that DQN achieves the lowest vehicle waiting time, DDQN exhibits unstable training behavior under the considered settings, and D3QN improves throughput at the cost of increased delay.

Keywords—Adaptive Traffic Signal Control, Deep Reinforcement Learning, Deep Q-Network (DQN), Double Deep Q-Network (DDQN), Dueling Deep Q-Network (D3QN), Markov Decision Process, SUMO, TraCI, Intelligent Transportation Systems, Traffic Optimization, Urban Traffic Control.

I. INTRODUCTION

Traffic congestion has become one of the most critical issues in modern cities due to the high urbanization rate and sustained increase in the number of privately owned vehicles [1]. The overloaded road systems will cause extreme congestions, travel time, fuel consumption and major environmental pollution. Signalised intersections are the major bottlenecks in particular where conflicting traffic flows need to have scarce resources of time and space and inefficient managed signals along these lines can significantly decline the overall functioning of the urban transportation systems. Conventional timing plans: Traffic signal control methods like fixed-time and actuated controllers are based on previously established timing plans or restricted sensor feedback and hence poorly respond to extremely dynamic and stochastic traffic [2]. Artificial Intelligence (AI) and reinforcement learning (RL) in particular have become the potential new trends in adaptive traffic signal control in recent years [3]. In contrast to supervised learning, that requires labelled data-sets, RL allows an agent to also learn optimal control policies by

interacting with the environment by trial and error. Deep Reinforcement Learning (DRL) procedures, which combine Deep Neural Networks (DNNs) with RL, have the potential to support the presence of high dimensionality and complexity in the state space of traffic states which makes them applicable in real-world problems with many lanes, traffic dynamics that are nonlinear and variable demand [4]. Some studies have documented that DRL based controllers may outperform traditional methods and heuristic methods in terms of lessening waiting time, length of queues and general congestion [5]. In spite of these developments, the current literature is mainly concerned with the development of new architectures or large-scale multi-intersection coordination structures. Relative lack of effort has been made to systematic and controlled empirical studies of closely related value-based DRL algorithms that run under the same experimental conditions, most notably in terms of learning stability, convergence behaviour and performance trade-offs. Additionally, indirectly many assume that the more elaborate architectures, e.g. Double DQN and Dueling DQN will always be better than the default DQN, but this may not be always true in the restricted conditions such as isolated intersections or limited training horizons. These observations will be the basis of this paper which will explore how deep reinforcement learning can be applied to achieve adaptive traffic signal control at an isolated intersection on the Simulation of Urban Mobility (SUMO) platform and the Traffic Control Interface (TraCI). The intersection control problem is presented as a Markov Decision Process and three popular value-based DRL algorithms, Deep Q -Network (DQN), Double Deep Q-Network (DDQN) and Dueling Deep Double Q-Network (D3QN) are designed and tested in the same simulation environment. This study will not offer a new learning algorithm, but an empirical comparison with respect to the traffic efficiency and learning behaviour between these methods. This work made some key contributions, which could be summarised as follows:

- i. SUMO and TraCI are used to establish a single DRL based traffic signal control framework, where DQN, DDQN and D3QN are tested with identical state representation, action space, reward function and traffic demand.

- ii. Multiple performance measures are conducted in order to give a systematic comparative analysis of those performance measures like total waiting time, average waiting time, throughput and training convergence behaviour.
- iii. The research provides empirical evidence of trade-offs between the complexity of architecture and control performance and learning stability, as well as a situation where other less complex models like DQN relative to more complex ones achieve better performance in narrow training care.

The rest of the paper is structured in the following way. Section II surveys the corresponding literature on the field of reinforcement learning-based traffic signal control. Section III outlines the proposed methodology that requires the Markov Decision Process formulation, the state space, the action space, the reward design and the learning architectures. Section IV includes an experimental setup, results and discussion. Last but not least, the paper ends with Section V, which summarizes the results and provides limitations and future research directions.

II. RELATED WORK

The last few years have been accompanied by tremendous advancement in applying deep reinforcement learning (DRL) to adaptive traffic signal control. In a broad overview, Panagiotis Michailidis, Iakovos Michailidis, Charalampos Rafail Lazaridis and Elias Kosmatopoulos point to the development of the reinforcement learning-based approaches to controlling traffic signals and provide an overview of the practical issues related to the implementation of this traffic control variant into the real world, including scalability and stability [6]. Neither DARPA-inspired original formulations nor those of Long Lv, Yang Liu and Fei-Yue Wang showed strong results on the ability to learn signal timing policies directly based on the states of traffic, though the latter formulations continue to be a popular reference point in the literature of value-based DARPA-inspired signal timing [7]. Newer works have aimed at making value-based methods more efficient and robust in their learning. To increase learning stability and performance in the adaptive signal control, Yongjie Zheng, Feng Zhang, Xiuming Xu, Yuanlin Liu and Yongjun Chen came up with Pri-DDQN that adds priority mechanisms to Double DQN [8]. Clement Cai, Yu Liu, Zhi Li, Feng Zhu and Qing Wang introduced a further deep reinforcement learning architecture which optimized the design of state and awards in order to attain better urban traffic signal operation in realistic simulation [9]. Equally, S. Mehmood Swapno, Mahub ur Rashid, Fatima Urooj and Mohammad Akram have shown the feasibility of Deep Q-Networks to reduce congestion and enhance traffic flow in traffic when the conditions are more complex [10]. The other training paradigm has been ventured into besides online learning. Lei Wang, Yu-Xuan Wang and Jian-Kang Li explored the offline reinforcement control of adaptive traffic signal control, which optimizes the efficiency of data and training stability in case of online exploration is scarce [11]. Under congested traffic conditions, H. Lee, Y. Cho and J. Kim studied the dynamics of reinforcement learning controllers and demonstrated the control of sensitive reward and state with regard to the stabilisation of performance [12]. Another research direction on the use of multi-agent-based

and network-level control is becoming a more active one. The concept behind the unification of multi-agent reinforcement learning towards traffic signal management suggested by Rohit Bokade and Xiaoning Jin as PyTSC allows the experimentation of a multi-jurisdictional situation with multiple intersections to be scaled [13]. A digital twin framework was used to optimize signal coordination in complex networks in which Vijayalakshmi Kumarasamy, Abhilasha Saroj, Yu Liang and Mina Sartipi combined decentralized graph-based multi-agent reinforcement learning [14]. Similarly, V. Krishna, S. Dommari and A. Reddy used the multi-agent deep reinforcement learning to control adaptive traffic signals in a smart city and also revealed the potential of distributed learning in large-scale implementations [15]. Various studies have also tackled the issue of heterogeneous traffic and region-specific issues. A paper by Agarwal, Priya Saxena and Rajesh Singh introduced a deep reinforcement learning framework of traffic signal control over a dynamic traffic in a heterogeneous traffic, which is especially applicable in the case of traffic situations in developing countries [16]. Taken together, these studies demonstrate that advanced architectures and multi-agent formulations may enhance throughput and scalability but, in most cases, it brings more training complexity and sensitivity to hyperparameters. Unlike the previous articles that mainly address network-level coordination or architecture improvements, the current study is based on a controlled and single-intersection environment and makes a methodical comparison between three weakly different value-based DRL frameworks: DQN, Double DQN and Dueling DQN. Through the examination of these approaches on the same simulation conditions and performance measures, the study will help to elucidate the trade-offs between architectural complexity, learning stability, delay reduction and throughput maximization on an adaptive traffic signal control.

III. METHODOLOGY

A. Markov Decision Process

The adaptive controlling traffic lights at an isolated intersection problem is formulated as a discrete time Markov Decision Process (MDP) with tuple of the form (S, A, P, R, γ) . In this case, S represents the state space; A represents the set of admissible actions; $P(s'|s, a)$ represents the conditional state-transition probability; $R(s, a, s')$ represents the reward function and $\gamma \in (0, 1)$ represents the geometric discount factor. On each discrete decision time, the decision-maker factors in the current state, s_t in S , chooses an action a_t in A , gets a scalar reward r_t and the system is self-evolved to the next state s_{t+1} . In the current architecture, transition kernel $P(s' \forall s, a)$ is not used, but rather a value of a microscopic simulator of the traffic, SUMO, providing its Traffic Control Interface (TraCI). When an action is performed, that is, the signal phase is selected the simulator runs a single control step calling *traci.simulationStep()* and the subsequent state is calculated out of the current traffic states. In this way, all the environmental dynamics, including the vehicle interactions, car-following dynamics and queue dynamics are fully

captured by SUMO and the learning agent is wrapped by a closed-loop interaction paradigm with this environment.

B. State, Action and Reward Design

1) State Space

The vector S_t is a representation of the state of the traffic at that intersection at a specific time t which is discrete in nature. In line with implementation that was taken up, the queue lengths and the total waiting times in each inbound lane constitute the state:

$$s_t = \left\{ \left(q_i^t, w_i^t \right) \mid i=1, 2, \dots, N_L \right\} \quad (1)$$

Whereby q_i^t and w_i^t are used to represent respectively the queue length and cumulative waiting time of lane i at epoch t and N_L is used to denote the number of incoming lanes. This definition provides both the information regarding the volume of traffic (in terms of queues) and the level of congestion (in terms of waiting time) to the agent.

2) Action Space

Action space is the set of admissible configurations of the traffic signals denoted by the set of possible configurations of the traffic signal, which is denoted by the set of actions, denoted as: A .

$$A = \{a_1, a_2, \dots, a_{N_p}\} \quad (2)$$

There is a phase of one of the prescribed signals that is associated with each action. The choice of a single phase at each control interval is made by the agent, with safety itself, such as having a minimum green duration and no forceful amber period, maintained by the logical of the signal-controller.

3) Reward Function

To motivate reduction in the congestion, the signal that can be used as a reward is the negative change in aggregate waiting time between consecutive time periods.

$$r_t = -(W_{t+1} - W_t) \quad (3)$$

Where the waiting time of all vehicles at time t is denoted by W_t . In turn, the effect of decreased general waiting time is rewarded and those actions which create more delays are punished; this aspect directs the learning process towards delay-reduction measures.

C. Deep Q-Learning Framework

Deep Q-Learning Framework Q-learning aims to approximate the optimal action-value function $Q^*(s, a)$ of finding the maximum expected discounted return in state s with action a then proceeding to the optimal policy. The Bellman optimality equation is as follows:

$$Q^i(s, a) = E \left[r_t + \gamma \max_{a'} Q^i(s', a') \mid s, a \right] \quad (4)$$

Deep Q-Networks Deep Q-Networks (DQN) approximate the action-value function with a neural network with parameters θ . $Q(s, a; \theta)$. The network is trained so as to minimize the mean squared error of the predicted Q-value and a target value:

$$L(\theta) = E_{(s, a, r, s') \sim D} \left[(y_t - Q(s, a; \theta))^2 \right] \quad (5)$$

In which the target y_t is calculated as:

$$y_t = r_t + \gamma \max_{a'} Q^{\bar{i}}(s', a')$$

and $\theta^{\bar{i}}$ are the parameters of a target network which are periodically updated with the online network. Sampling mini-batches of a replay buffer D is used to destabilize training by experience replay.

D. Double Deep Q-Network (DDQN)

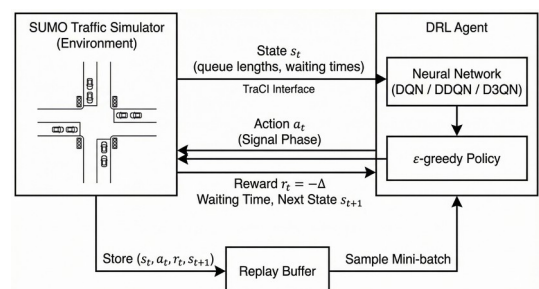
The use of max operator in the target calculation of Standard DQN has made the algorithm susceptible to overestimation bias. The solution to this problem by Double Deep Q-Network (DDQN) is the decoupling of action selection and action evaluation. In DDQN, the online network with parameters θ is applied to choose the optimal action and the target network with parameters $\theta^{\bar{i}}$ is applied to assess its value:

$$y_t^{DDQN} = r_t + \gamma Q^{\bar{i}}(s', a')$$

This adaptation minimizes the optimistic bias in the estimation of values and in most cases results in a more stable learning behavior in stochastic systems like urban traffic systems.

E. Dueling Double Deep Q-Network (D3QN)

The dueling architecture breaks down the action-value function into a state-value function $V(s)$ and an advantage function $A(s, a)$. The network will then be divided into two streams, one of which approximates $V(s)$ and the other approximates $A(s, a)$. They are added up to get the final Q-values as:



$$Q(s, a) = V(s) + \left(A(s, a) - \frac{1}{|A|} \sum_a A(s, a') \right) \quad (8)$$

This definition is identifiable and can learn the relative significance of states and actions more effectively.

F. System Architecture and Learning Workflow

The general system architecture and learning workflow are shown in Fig. 1.

Fig. 1. System architecture and training workflow of the proposed deep reinforcement learning-based adaptive traffic signal control framework using SUMO and TraCI.

The environment in this case is the SUMO traffic simulator which is connected to the learning agent at the Traffic Control Interface (TraCI). At every control stage, the agent gets the current state of traffic (length of queues and waiting times) in SUMO. According to this state, the agent chooses a signal phase based on the ϵ -greedy policy and applies the action to traffic signal controller. This is followed by the simulator moving the environment by one step and the reward and the next state is received. Every transition tuple (s_t, a_t, r_t, s_{t+1}) is stored in replay buffer and used to update the neural network parameters through mini-batch gradient descent. This feedback loop keeps interacting with this process in a recurring manner in every episode until the agent finally learns to effectively control signals policy. DQN, DDQN and D3QN have the same workflow and are compared fairly and in a controlled environment.

G. Network Architecture and Training Details

All the neural networks deployed in DQN, DDQN and D3QN are fully connected feed-forward models, as shown in Fig. 2.

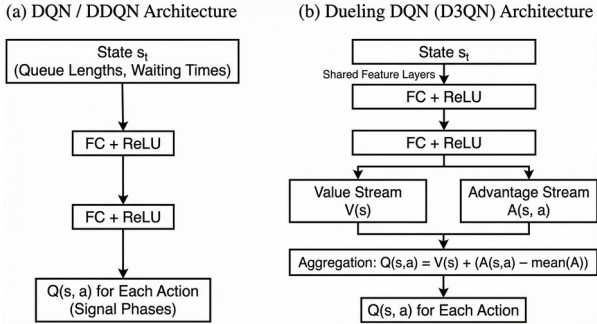


Fig. 2. Neural network architectures for (a) DQN/DDQN and (b) Dueling DQN (D3QN) used for action-value function approximation.

The input layer obtains the state vector that consists of queue length and waiting time of each of the incoming lanes. This is then succeeded by two concealed layers of ReLU activation functions which carry out nonlinear feature extraction. One Q-value is generated by the output layer corresponding to each possible action (signal phase). In the case of DDQN, both online and target networks are set up in the same architecture but with different parameters. In the case of D3QN, the network is divided into streams of values and advantage following the shared layers and combined to derive the final Q-values. The networks are trained on Adam optimizer and mean squared error loss. Experience tuples

(s_t, a_t, r_t, s_{t+1}) are stored in a replay buffer of size 100,000 transitions. Training is done with mini-batches that have a size of 512. The discount factor is set to $\gamma = 0.99$. In the case of DDQN and D3QN, a target network is upheld and changed regularly every fixed amount of training steps to enhance stability. An ϵ -greedy philosophy of exploration is followed, in which the agent chooses an action chosen randomly with probability ϵ and the greedy action with the remaining probability. During the implementation, ϵ is set to 0.1, decays with $5 * 10^{-4}$ per step and limited to a lower value of 0.05 in order to keep on exploring.

IV. RESULTS AND DISCUSSION

A. Experimental Setup and System Requirements

The suggested deep reinforcement learning-based traffic signal control model was tested in the Simulation of Urban Mobility (SUMO) simulator with Time Traffic Control Interface (TraCI) interface. PyTorch deep learning library was the Python implementation of the learning agents. The whole experimentation was performed on a normal desktop computer with a multi-core processor with a graphic card where applicable, having a 64-bit operating system. A predetermined SUMO network and route file was used to configure the simulation environment and each experiment was run over a fixed number of simulation steps per episode through several training epochs. DQN, DDQN and D3QN all were using identical traffic demand, network configuration and training settings to provide a fair and controlled comparison. All methods had the same learning parameters, with the discount factor, replay buffer size, batch size and exploration schedule. Each episode was completed by a reset of the simulator, then the agent was connected to the environment via TraCI, by observing the prevailing traffic condition, choosing a signal phase and obtaining a reward according to the resulting conditions in traffic. The reported outcomes are related to the performance of convergence after the training, by using the model that is performing the best at the time of saving it in the process of training.

B. Evaluation Metrics

In order to have a holistic and interpretable performance in traffic control, four evaluation criteria were taken into consideration.

1) Total Waiting Time (ms)

This parameter is a sum of the waiting time in all the lanes at all simulation steps in an episode. SUMO records waiting time in seconds, but to make it coherent and understandable all the values of waiting time are multiplied by a thousand and suggested in milliseconds (ms). The measure is a standard congestion cost across the globe with lower scores registered as more efficient concerning traffic.

2) Average Waiting Time (ms)

The average waiting time is calculated as the ratio of cumulative total waiting time and the total figure of the vehicles which passed through the network throughout the episode. This gives an approximation value of delay per-vehicle and is a more interpretable value of user-level traffic performance.

3) Throughput (vehicles)

The throughput is specified as the count of vehicles to pass the network or trucks that made it through the network in the

episode of the simulation. The presence of higher throughput suggests that the road network is utilized better and the traffic flow is improved.

4) Average Queue Length (vehicles)

The average length of the queue is calculated to be an average of time based the number of stalled vehicles per all controlled lanes in the entire episode. This metric shows the average level of congestion at the intersection and is a complement to that of the waiting time.

All measures are calculated in a similar manner across the three methods so as to give justifiable comparison.

C. Quantitative Performance Comparison

Table I summarizes the comparative performances of the three reinforcement learning models.

TABLE I. COMPARATIVE PERFORMANCE METRICS OF DRL MODELS

Metric	DRL Models		
	DQN	DDQN	D3QN
Total Waiting Time (ms)	1,217,610	6,098,928	4,789,749
Avg. Waiting Time (ms)	1,036.26	17,626.96	2,698.45
Throughput (Vehicles)	1,175	346	1,775
Avg. Queue Length (vehicles)	167	179	188

The outcomes show that the controller based on DQN reduces the overall waiting time as well as the average waiting time of the three approaches, which proves that it works well in minimizing the overall traffic delay under the simulation conditions in question. Contrarily, DDQN has much greater values of waiting time and much less throughput, which indicates unsteady or ineffective learning behaviour in the case at hand. D3QN has the greatest throughput meaning that it can accommodate more vehicles to cross the intersection at the end of the same time interval in the simulation. Yet, such an increase in throughput is at the expense of higher total and average waiting times than DQN. This observation is further supported by the average queue length values with DQN having the minimum average queue length and the value of D3QN having greater average queue length in response to the trade-off between the flow and minimizing the delay. It is also worth mentioning, that absolute values of the waiting time will depend on the traffic demand that is congested, as well as on the cumulative character of the measurement. Hence, the main area of concern in this analysis is the differences in the performance of the three techniques in the same conditions of the experiment.

D. Training Convergence Analysis

Each model was further investigated into the learning behavior by converging the training plots through convergence training plots, where the total waiting time was tracked as the epoch progressed.

1) DQN Training Behaviour

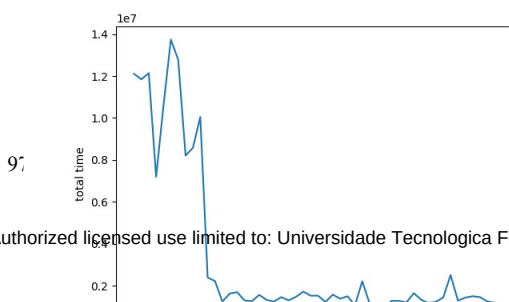


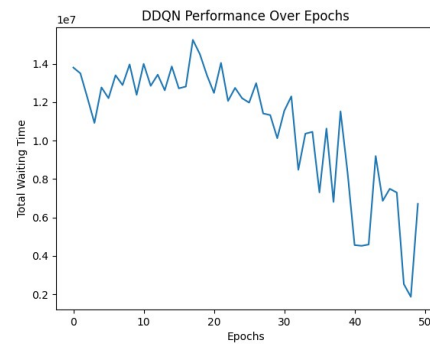
Fig. 3. Training convergence of the Deep Q-Network (DQN) model

Fig. 3. The DQN model exhibited a high convergence rate in the first few training epochs. There is a significant reduction in the waiting time in the early exploration phase and it then levels off to a low value, thus representing a healthy and stable learning trajectory. This is an L-shaped converging trajectory, which is an indicator of effective exploration followed by the permanent exploitation of the obtained policies.

2) DDQN Training Behaviour

Fig. 4. The DDQN convergence plot has a strong set of oscillations and variance. There is no continuous downward trend, which assumes instability in learning and difficulties in refining policy representations. These changes support the quantitative performance decline that is reported in Table I.

Fig. 4



Training convergence of the Double Deep Q-Network (DDQN) model

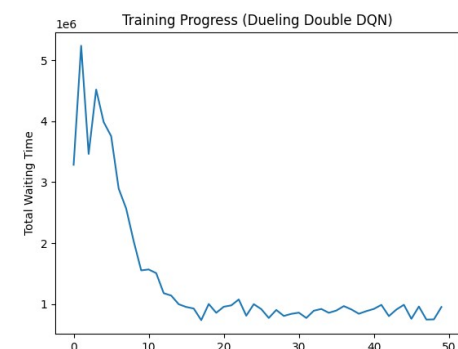
3) D3QN Training Behaviour

D3QN model (Fig. 5.) illustrates the values of waiting time decreasing gradually, which indicates the process of progressive learning; however, the convergence rate is relatively low compared with that of DQN. The complexity of the network also adds new parameters, requiring more training samples to achieve fine-grained optimization.

Fig. 5. Training convergence of the Dueling Double Deep Q-Network (D3QN) model

E. Discussion and Performance Trade-offs

The findings of the experiment indicate a trade-off that exists between the minimization of delays and the maximization of throughput. DQN will cause the minimal cumulative and average waiting times and the minimal average queue length and therefore, will perform better on congestion reduction



and user-level delay. D3QN, on the other hand, has the most throughput, implying that it is more focused on the large number of vehicles moving at the intersection at the expense of long wait times and long queues. These facts suggest that an algorithm should be selected based on a particular operational goal of a traffic management system. DQN seems to be the most appropriate option given the circumstances discussed in the case of the primary aim at reducing delay and queue formation. In the event that maximizing throughput is the overriding goal, however, D3QN could prove of benefit, at the expense of increased delay. The low result of DDQN in the paper suggests that additional attention to hyperparameters or extended training periods can be needed to enable this system to achieve the theoretical advantages of this approach in practical traffic control contexts. Lastly, it should be stressed that the results obtained are reported on one intersection with a constant traffic demand pattern. Although the results offer some valuable clues about the comparative performance of DQN, DDQN and D3QN, additional experiments with multiple random seeds, traffic density variations and more elaborated network structures would be required to make more general conclusions on the aspect of robustness and scalability.

CONCLUSION

This work introduced a deep reinforcement-based adaptive framework of traffic signal control to overcome the drawbacks of the traditional fixed-time and rule-based approaches. The intersection control problem has been modelled as a Markov Decision Process and estimated on the basis of realistic-simulation environment. Three distinct deep reinforcement learning architectures, which consist of DQN, Double DQN and Dueling Double DQN, were put into practice and compared under the same conditions. The findings further suggest that the standard DQN has the lowest average waiting time and has more stable training behaviour in the taken single-intersection scenario. Even though Double DQN and Dueling Double DQN have theoretical benefits, being more complex in architecture with increased sensitivity to training conditions, both are slower and less predictable in learning within such short training periods, whereas DQN is proven to be effective and more reliable in learning within such short training ranges. These results indicate that a more complex model is not always associated with improved performance associated with isolated intersection control. This paper is confined to one intersection and a given traffic demand, thus future research will involve expanding the framework to multi-intersections and multi-corridor networks with multi-agent reinforcement finding, consideration of strong robustness in terms of noisy and incomplete sensing agendas and multi-objective objectives like the priority of emergency vehicles, preference of the public transport mode, fuel consumption and minimizing of emissions through

multi-objective learning. Moreover, real-time implementation on embedded systems like NVIDIA Jetson will also be explored to measure latency, stability and safety in an effort to balance the difference between simulated evaluation and real-world implementation of intelligent traffic management systems.

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